

Classical Molecular Dynamic Simulation of Liquid Water

NITIN SHARMA

1 Introduction

Water plays an important role in life and in everyday processes. It has applications in almost every field, including biological and industrial fields, among others. Despite being a simple molecule, it is remarkably difficult to model its interactions accurately. It is easier to understand the intermolecular potential in an assembly of noble gases than in the same number of atoms inside water.

Water is a universal solvent because it dissolves substances more easily than any other liquid. So, how this dielectric constant (i.e., how charges separate and make it polar) works for water is another challenge to study, as is how it varies with different force-field parameters. Computational studies are very useful for investigating variations in system size, temperature, atomic interactions, and many other factors that cannot be investigated experimentally.

Water simulation was reported in the 1960s [1, 2], but the first generation of water models, which are widely used today, was developed in the 1980s. The simple point charge (SPC) water model, published by Berendsen et al. [3] in 1981, is based on the tetrahedral structure of ice; the force field used is the GROningen MOlecular Simulation (GROMOS) [4]. The Transferable Interaction Potential 3-point (TIP3P) model, published by Jorgensen et al. in 1983 [5], is based on gas-phase water; the force field used is Assisted Model Building with Energy Refinement (AMBER) [6] and Optimized Potentials for Liquid Simulations (OPLS) [7]. In that same paper, Jorgensen et al. also mention a 4-point model, TIP4P, in which the negative charge of oxygen was removed from the LJ potential, thereby improving structural properties [5]. This model is mainly used with the OPLS force field. Then, in 2000, the 5-point model, TIP5P, published by Mahoney et al. [8], reproduces the density of water as a function of temperature (-37 to 100 °C). Furthermore, there are many modifications to these 3-, 4-, and 5-point models to improve properties using new computational techniques or to reduce computational costs. These computational simulations of water started with the pioneering work of Barker and Watts, followed by Rahman and Stillinger using the Ben-Naim Stillinger potential and proposing the ST2 potential after a few years. In these studies, classical statistical mechanics is used with a simple rigid, non-polarisable model to describe water interactions. Since then, many models have been proposed, including rigid and flexible models, as well as polarisable and non-polarisable models. It is still quite difficult to understand how the structural and dynamic properties behave under different force parameters. There are many models available, but researchers aim to find one that accurately reproduces all experimental values.

The objective of this study is to examine the effect of different models on the structure and dynamics of water at different temperatures. Classical MD simulation of different water models and at different temperatures is conducted and computed following properties: density, dielectric constant, radial distribution function (RDF), hydrogen bond analysis, and diffusion coefficient. The next section describes the computational methodology, and the subsequent section provides the Results and Discussion with a summary at the end.

2 Computational Methodology

The optimized geometry of water was obtained using Gaussian 09 software [9]. A cubic box of ≈ 8.1 nm is prepared using optimized water molecules, and it is then replicated to 500-5000 molecules. All the classical MD simulations were performed using GROMACS-5.0.7 software [10]. For water molecules, the bonded and non-bonded parameters were taken from the OPLS-AA force field [7] and from different water models (SPC, TIP3P, SPC/E, and TIP4P).

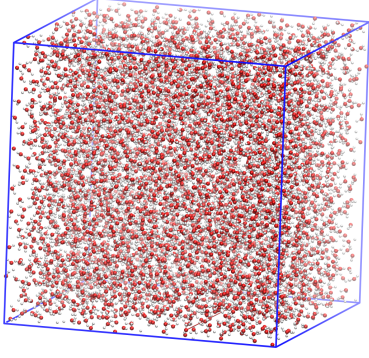


Figure 1: Snapshot of simulated water box at 298 K. Color scheme: Red (sphere)–Oxygen, white (sphere)–Hydrogen.

The force field parameters for different water models are given in **Table 1**. and the Kadaoluwa et al. [11] and Morozova et al [12]. interaction parameters. The cutoff for the non-bonded interaction was fixed to 1.2 nm. All Newton’s equations were solved using the velocity-verlet algorithm [13] with a time step of 1 fs. The steepest-descent algorithm [14] is used to minimize the energy, followed by an NPT run for 10 ns using the Parrinello-Rahman barostat [15] and the Nose-Hoover thermostat [16]. Finally, a 25 ns production run (NVT ensemble) has been performed using equilibrated geometry, and the calculated average density across models deviates by $< 1\%$ from the literature survey at 298 K. The MD snapshot of the water box is shown in Figure 1(See **Fig 1**).

Table 1: Force Field Parameters for Water Models [11]

Model	O σ (Å)	O ϵ (kcal/mol)	O charge (e)	M charge	H charge (e)	O–H bond (Å)	H–O–H angle (deg)	O–M bond (Å)
SPC	3.166	0.15535	-0.82	–	0.41	1.0	109.47	–
TIP3P	3.15061	0.1521	-0.834	–	0.417	0.9572	104.52	–
SPC/E	3.166	0.15535	-0.8476	–	0.4238	1.0	109.47	–
TIP4P	3.15365	0.155	–	-1.04	0.52	0.9572	104.52	0.15

3 Results and Discussion

3.1 Effect of system size

We have 4 different system sizes with 500, 2000, 5000, and 10000 water molecules to examine density deviations as a function of system size. As the system size increases, the density approaches the experimental value (See **Table 2**). However, a saturation point in density reached after 5000 molecules, so further with 5000 water molecules were proceeded with different water models. Our density obtained for SPC at 298 K for 5000 molecules is 989.241.

Table 2: Density (kg/cm³) of SPC/E Water Model at 298 K for Different System Sizes

No. of Molecules	Density	No. of Molecules	Density
500	987.26	5000	989.241
2000	988.723	10000	989.325
Experimental		997[11]	

3.2 Bulk Properties

3.2.1 Density

Density is defined as the mass of a substance per unit volume. Density is calculated by dividing the box into small slices and analyzing how the particles distribute over time. Model TIP4P, SPC/E shows a very good value of density (near the experimental, with dev. $< 1\%$) (See **Table 3**). At the same time, the other two models show deviations exceeding 2%. Density decreases with increasing temperature, indicating thermal expansion of the liquid.

3.2.2 Dielectric constant

The static dielectric constant(ϵ) was calculated using fluctuations of the total dipole moment and can be written as:

$$\epsilon = 1 + \frac{\langle M^2 \rangle - \langle M \rangle^2}{3\epsilon_0 V k_B T} \quad (1)$$

where ϵ is the permittivity of free space, M is total dipole moment, V is the volume of the box, k_b is boltzmann constant and T is temperature. The SPC/E model can produce a better value (See **Table 3**). We observe that as the temperature increases, ϵ decreases, which is consistent with experimental measurements [17].

3.3 Solvation Shell Analysis

Different interactions between water molecules are described using radial distribution function (RDF), and can be written as,

$$g_{ab}(r) = \frac{V}{N_a N_b} \sum_{i=1}^{N_a} \sum_{j=i+1}^{N_b} \langle \delta(r - |\vec{r}_i(t) - \vec{r}_j(t)|) \rangle_t \quad (2)$$

where N_a and N_b are the number of particles of type a and b , respectively, V is the volume of the simulation box, and $\vec{r}_i$ and $\vec{r}_j$ are the positions of particles of type a and b , respectively [18]. Coordination number(CN) of water is calculated using the first minimum in the RDF plot, and can be written as [18],

$$CN = 4\pi\rho \int_0^r g(r) r^2 dr \quad (3)$$

where (r) is the distance up to which the coordination number is evaluated, ρ is the density of particles.

3.3.1 Effect of water model

For the Ow-Hw RDFs, the SPC model shows a first-peak maximum at 0.180 nm at 298 K. Whereas, SPC/E, TIP3P, and TIP4P models show the first maxima at 0.176 nm, 0.184 nm, and 0.182 nm, respectively (see **Figure 2a**). The experimentally determined Ow-Hw distance, obtained using (write down the method), is 0.173 nm [19], and among these models, SPC/E yielded the closest value. These findings highlight both similarities and differences among the water models. For the Ow-Ow RDFs, the SPC/E and TIP4P models show the maxima at 0.274 nm, while SPC and TIP3P show it at 0.278 nm (see **Figure 2b**), comparable to the experimental value of 0.275 nm [20].

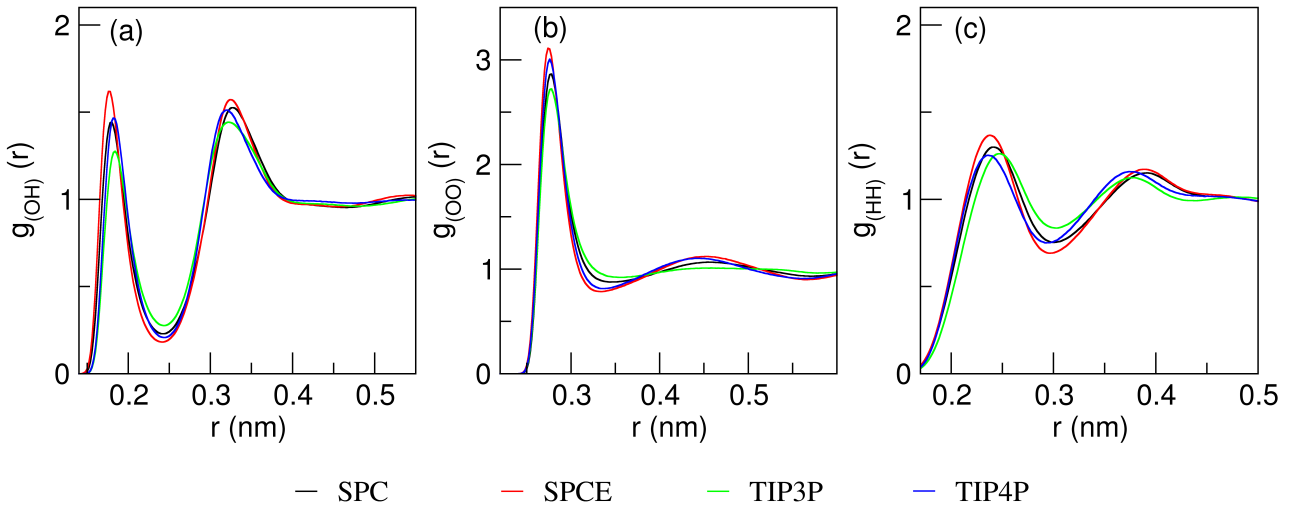


Figure 2: RDF of (a) O-H, (b) O-O, (c) H-H for different models at 298 K temperature

In terms of $g_{HH}(r)$, at 298 K, the SPC model peaks at 0.242 nm, whereas SPC/E, TIP4P, and TIP3P peak at 0.238 nm, 0.238 nm, and 0.246 nm, respectively (see **Figure 2c**), aligned with the experimental observation (0.238 nm) [20]. The CN for SPC, SPC/E, TIP3P, and TIP4P are 5.32, 4.77, 5.94, and 5.06, respectively, whereas the experimental data match this at 4.7 (See **Table 3**).

3.3.2 Effect of Temperature

In the Ow-Hw radial distribution function (RDF), the peak intensity for the SPC/E model is significantly higher and is shifted more to the left, while the TIP3P model exhibits the lowest peak intensity and is shifted to the right side of the graph (See **Figure 5-7** from the index). This difference occurs because the SPC/E model is more ordered than the others. The other two RDFs, namely OW-OW and HW-HW, show similar patterns.

Additionally, the coordination number is influenced by temperature; as temperature increases, it changes.

Table 3: Comparison of structural and dynamic properties of Water Models with Different Temperatures

Temp (K)	Model	Density (kg/m ³)	Coordination no.	Dielectric constant	H-Bonds per molecule	Self-Diffusion ($\times 10^{-5}$ cm ² /s)
298	SPC	967.047	5.32	65.361	3.457	4.3507 $\pm$ 0.0503
	SPC/E	989.241	4.77	70.8847	3.594	2.6009 $\pm$ 0.0257
	TIP3P	974.808	5.94	97.9297	3.337	5.9088 $\pm$ 0.0546
	TIP4P	993.807	5.06	51.8287	3.549	3.5999 $\pm$ 0.1027
320	SPC	949.777	5.66	61.4641	3.341	5.9857 $\pm$ 0.0452
	SPC/E	976.584	4.99	69.4682	3.495	4.1377 $\pm$ 0.1081
	TIP3P	954.148	6.1	91.0317	3.218	7.8701 $\pm$ 0.0998
	TIP4P	979.595	5.29	50.4766	3.437	5.4662 $\pm$ 0.0875
340	SPC	931.577	5.89	59.9624	3.228	8.0608 $\pm$ 0.0990
	SPC/E	962.369	5.20	66.8455	3.399	5.5547 $\pm$ 0.0176
	TIP3P	932.98	6.62	82.699	3.108	9.8986 $\pm$ 0.3139
	TIP4P	963.571	5.51	49.2146	3.332	7.3002 $\pm$ 0.1638
360	SPC	911.374	6.10	56.9843	3.121	10.1703 $\pm$ 0.0391
	SPC/E	946.294	5.39	64.605	3.298	7.2690 $\pm$ 0.1519
	TIP3P	909.981	6.77	76.5731	2.991	12.2776 $\pm$ 0.0819
	TIP4P	944.988	5.73	48.1109	3.223	9.2321 $\pm$ 0.0233
Experimental		997	4.7	78.3	3.5-4.0	2.30

3.4 Hydrogen bonding

A hydrogen bond is a strong intermolecular interaction formed between the oxygen atom of a water molecule and the hydrogen atom of a neighboring water molecule. Molecular dynamics simulations were used to analyze hydrogen-bond formation statistics, including the average number of hydrogen bonds per molecule and the hydrogen-bond length.

The average hydrogen-bond distance remains nearly constant for various models at $\approx$ **0.2775 nm** within the temperature range. The number of hydrogen bonds describes the Oxygen-hydrogen interactions within a certain distance. The highest value of the number of hydrogen per molecule is for SPC/E, i.e., 3.594, while the lowest is 3.337 for TIP3P at 298K, which matches the experimental value of 3.5-4.0. We also observe that as the temperature increases, the number of hydrogen bonds decreases.

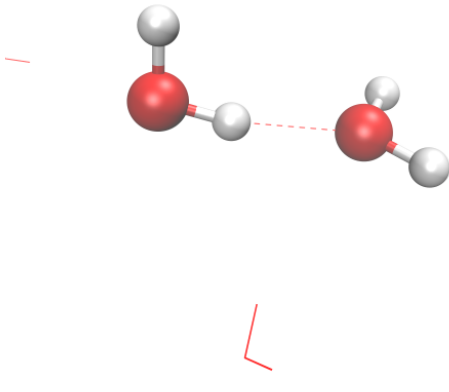


Figure 3: Hydrogen bond interaction between oxygen and hydrogen atoms.

The average number of hydrogen bonds per molecule, as presented in **Table 3**. Experimentally, liquid water contains approximately $\approx 3.5\text{--}4.0$ hydrogen bonds per molecule. The hydrogen-bond lifetime in bulk water is typically **5–10 ps**; therefore, trajectories saved every **1 fs** provide sufficient temporal resolution for accurate lifetime calculations and robust statistical analysis.

For any specific hydrogen bond i , a binary existence variable is defined as:

$$s_i(t) = \begin{cases} 1, & \text{if the H-bond exists at time } t \\ 0, & \text{if the H-bond does not exist} \end{cases} \quad (4)$$

GROMACS computes the average autocorrelation function over all hydrogen bonds:

$$C(t) = \frac{\langle s_i(t_0) s_i(t_0 + t) \rangle}{\langle s_i(t_0)^2 \rangle} \quad (5)$$

A rough estimate is calculated as the time integral of the autocorrelation function:

$$\tau_{\text{HB}} = \int_0^\infty C(t) dt \quad (6)$$

Using trAVIS (Trajectory Analyzer and Visualizer) [21], we calculate this hydrogen-bond lifetime for different models at 298K. The longest hydrogen-bond lifetime is with the SPC/E model, and the shortest

Table 4: Hydrogen bond lifetimes (ps) for different water models

SPC	SPC/E	TIP3P	TIP4P
2.4650	4.0142	1.7188	0.5883

with TIP4P. We also plot this hydrogen autocorrelation function (See **Fig 8**) in the index.

3.5 Diffusion Coefficient

The mobility of water molecules is calculated from the mean-squared displacement (MSD) vs time plots. The self-diffusion coefficients, D_i , are calculated from the linear regime of the MSD using Einstein’s relationship. The equation used to calculate D is stated below:

$$\lim_{t \rightarrow \infty} \left\langle \|\mathbf{r}_i(t) - \mathbf{r}_i(0)\|^2 \right\rangle_{i \in A} = 6D_A t \quad (7)$$

The calculated β values of the linear region lie between 0.99 and 1.01; here, the scaling exponent (β) is the slope of the dlogMSD vs dlogt plot, and can be written as,

$$\beta = \frac{d \log(\text{MSD}(t))}{d \log(t)} \quad (8)$$

We show different β values in **Table 4** in the index portion.

Using computational calculation, at 298 K, the closest value of Self-diffusion is of SPC/E, i.e., $2.6 (\times 10^{-5} \text{ cm}^2/\text{s})$ while the experimental value is $2.3 (\times 10^{-5} \text{ cm}^2/\text{s})$ and literature value is $2.60 \pm 0.03 (\times 10^{-5} \text{ cm}^2/\text{s})$ [11]. Moreover, as we increase the temperature, the mean square displacement also increases.

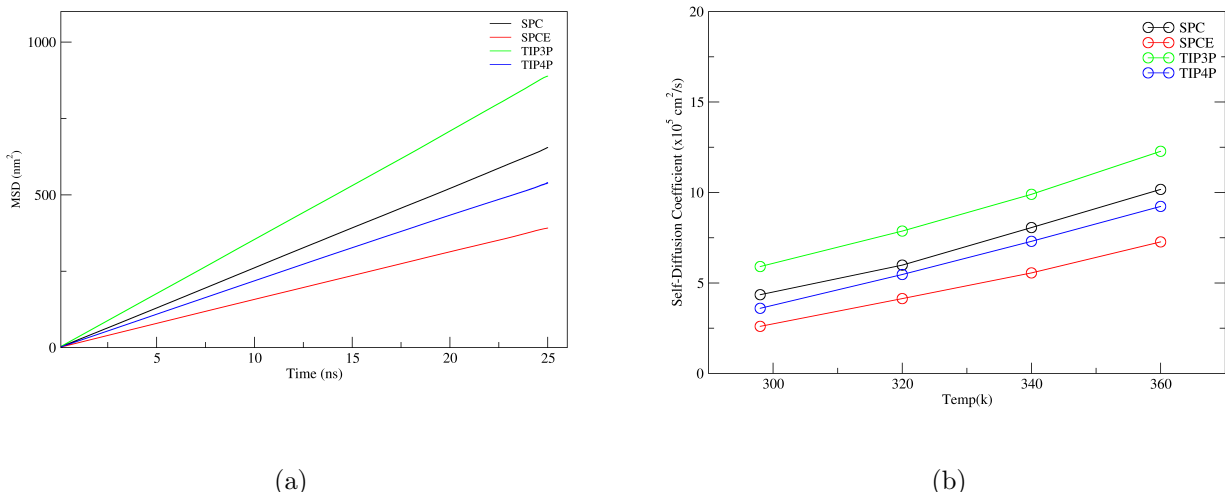


Figure 4: (a)Mean square displacement (MSD) at 298 K, (b) Diffusion coefficient vs temperature

4 Conclusion

This investigation explores the effects of different models and temperatures on the structure and dynamics of liquid water using Classical MD Simulations. The properties examined are Density, RDF, Coordination number, Dielectric constant, H-bonding, and Self-Diffusion coefficient. Density shows less than 1% deviation from the experimental value. Similarly, RDF's first maxima peak is also relevant to the experimental data. Furthermore, the dielectric constant also matches the experimental measurements; with increasing temperature, it decreases, validating the results. Moreover, we have Self-Diffusion showing a slight deviation from the experimental value; we can say that the four models taken cannot reproduce the exact value as the experiment may, or some others can. We cannot conclude which model can reproduce all properties, i.e., structural or dynamic, because each model has its own applications. However, we can say that SPC/E will be the best model among the four, as it produces a value close to the experimental one. We also observe a different temperature trend in properties, which is also validated by reported experimental measurements. We observe that density, coordination number, and self-diffusion coefficient increase with temperature, whereas hydrogen-bond per molecule and dielectric constant decrease with temperature.

5 Acknowledgement

I would like to express my sincere gratitude to Prof. Arun Venkatnathan for his valuable guidance, encouragement, and support throughout the course of this summer project. His mentorship greatly contributed to the successful completion of this work. I am equally thankful to my lab-mates, Nayan Dey and Vishnu Sudarshan, for their constant guidance, patience, and willingness to help me at every stage of this project. I would also like to extend my heartfelt thanks to Ms. Nisha Kurkure for her technical support, which was instrumental in helping me overcome various challenges during the project. I am grateful to IISER Pune for providing me with the opportunity, infrastructure, and resources needed to carry out this summer project.

References

- (1) Jorgensen, W. L.; Tirado-Rives, J. *Proceedings of the National Academy of Sciences* **2005**, *102*, 6665–6670.
- (2) Ouyang, J. F.; Bettens, R. P. *Chimia* **2015**, *69*, 104–104.

- (3) Berendsen, H. J.; Postma, J. P.; van Gunsteren, W. F.; Hermans, J. In *Intermolecular forces: proceedings of the fourteenth Jerusalem symposium on quantum chemistry and biochemistry held in jerusalem, israel, april 13–16, 1981*, 1981, pp 331–342.
- (4) Schmid, N.; Eichenberger, A. P.; Choutko, A.; Riniker, S.; Winger, M.; Mark, A. E.; Van Gunsteren, W. F. *European biophysics journal* **2011**, *40*, 843–856.
- (5) Jorgensen, W. L.; Chandrasekhar, J.; Madura, J. D.; Impey, R. W.; Klein, M. L. *The Journal of chemical physics* **1983**, *79*, 926–935.
- (6) Cornell, W. D.; Cieplak, P.; Bayly, C. I.; Gould, I. R.; Merz, K. M.; Ferguson, D. M.; Spellmeyer, D. C.; Fox, T.; Caldwell, J. W.; Kollman, P. A. *Journal of the American Chemical Society* **1995**, *117*, 5179–5197.
- (7) Jorgensen, W. L.; Maxwell, D. S.; Tirado-Rives, J. *Journal of the american chemical society* **1996**, *118*, 11225–11236.
- (8) Mahoney, M. W.; Jorgensen, W. L. *The Journal of chemical physics* **2000**, *112*, 8910–8922.
- (9) Frisch, M. Gaussian 09, Revision D. 01/Gaussian, 2009.
- (10) Abraham, M. J.; Murtola, T.; Schulz, R.; Páll, S.; Smith, J. C.; Hess, B.; Lindahl, E. *SoftwareX* **2015**, *1*, 19–25.
- (11) Kadaoluwa Pathirannahalage, S. P.; Meftahi, N.; Elbourne, A.; Weiss, A. C.; McConville, C. F.; Padua, A.; Winkler, D. A.; Costa Gomes, M.; Greaves, T. L.; Le, T. C., et al. *Journal of Chemical Information and Modeling* **2021**, *61*, 4521–4536.
- (12) Morozova, T. I.; García, N. A.; Barrat, J.-L. *The Journal of Chemical Physics* **2022**, *156*.
- (13) Verlet, L. *Physical review* **1967**, *159*, 98.
- (14) Payne, M. C.; Teter, M. P.; Allan, D. C.; Arias, T.; Joannopoulos, a. J. *Reviews of modern physics* **1992**, *64*, 1045.
- (15) Parrinello, M.; Rahman, A. *The Journal of Chemical Physics* **1982**, *76*, 2662–2666.
- (16) Nosé, S. *The Journal of chemical physics* **1984**, *81*, 511–519.
- (17) Malmberg, C. G.; Maryott, A. A. *Journal of research of the National Bureau of Standards* **1956**, *56*, 1–8.
- (18) Allen, M. P.; Tildesley, D. J., *Computer simulation of liquids*; Oxford university press: 2017.
- (19) Skinner, L. B.; Huang, C.; Schlesinger, D.; Pettersson, L. G.; Nilsson, A.; Benmore, C. J. *The Journal of chemical physics* **2013**, *138*.
- (20) Soper, A.; Benmore, C. *Physical review letters* **2008**, *101*, 065502.
- (21) Brehm, M.; Thomas, M.; Gehrke, S.; Kirchner, B. *The Journal of chemical physics* **2020**, *152*.

Index

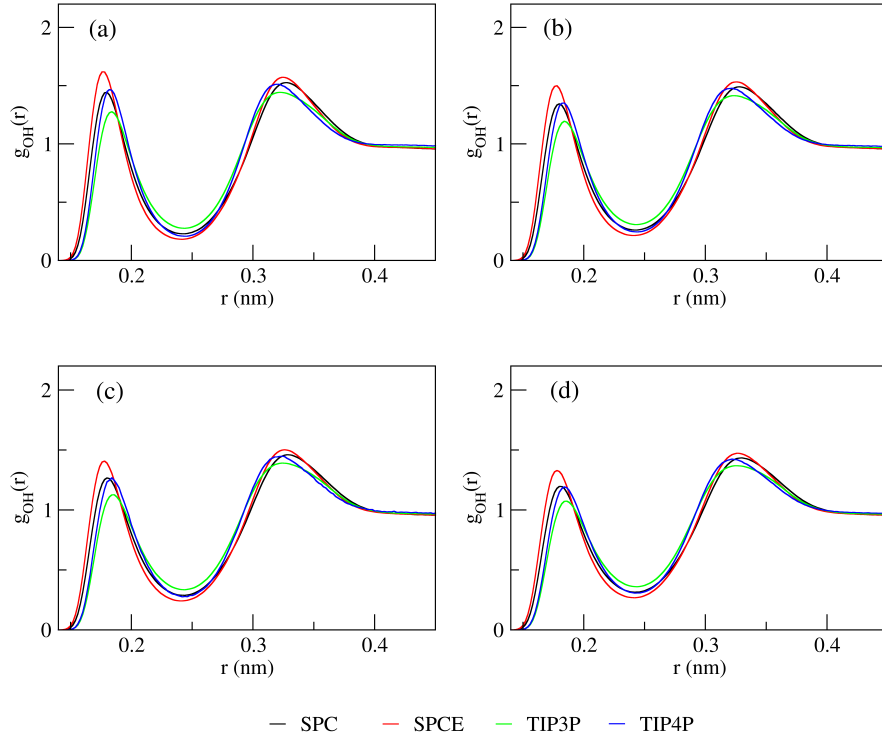


Figure 5: RDF of O-H for different models at (a) 298 K, (b) 320 K, (c) 340 K, (d) 360 K temperatures

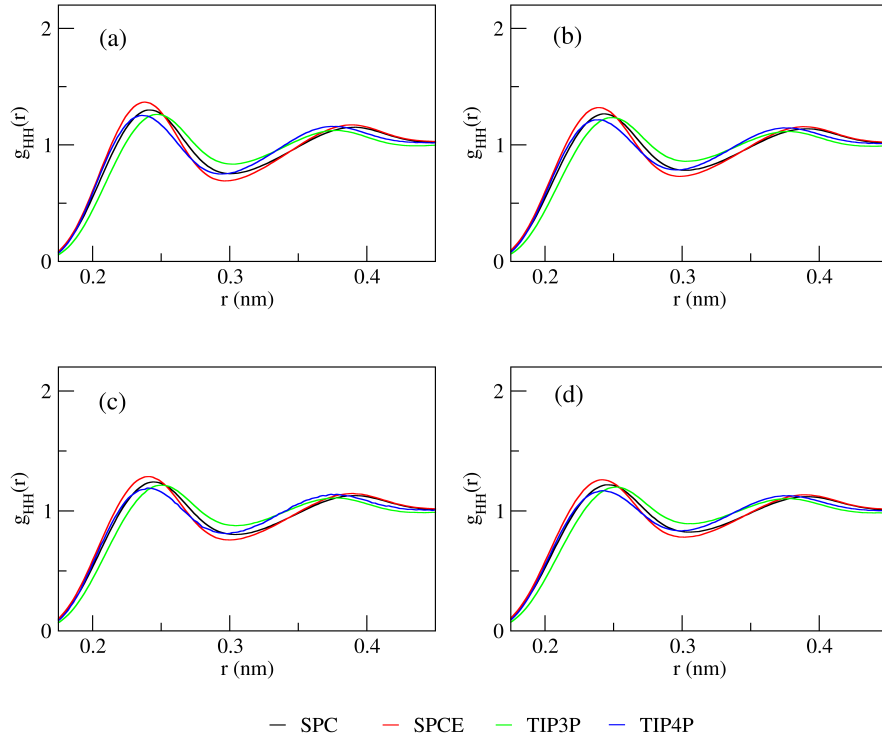


Figure 6: RDF of H-H for different models at (a) 298 K, (b) 320 K, (c) 340 K, (d) 360 K temperatures

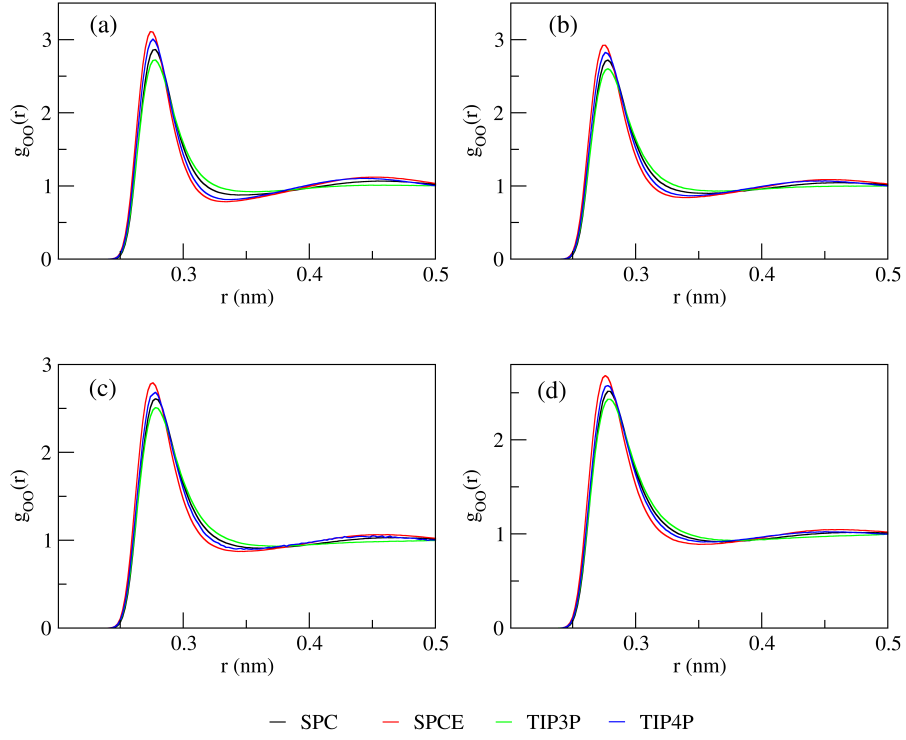


Figure 7: RDF of O-O interactions for different models at (a) 298 K, (b) 320 K, (c) 340 K, (d) 360 K temperatures

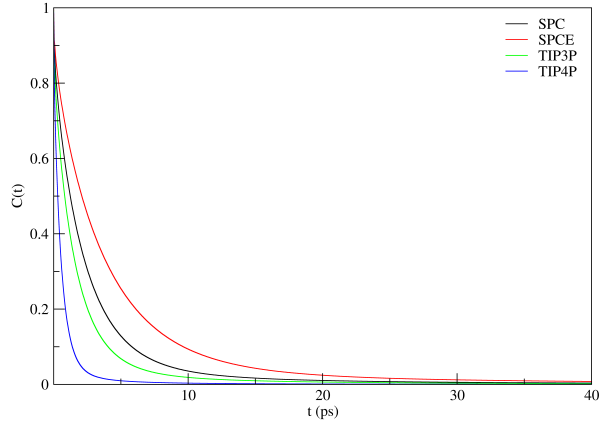


Figure 8: Hydrogen bond autocorrelation function, $C(\tau)$, as a function of time delay τ (ps) for different water models (SPC, SPC/E, TIP3P, and TIP4P).

Model	SPC	SPC/E	TIP3P	TIP4P
β	1.002	0.993	1.000	0.997

Table 5: Log-log slope (β) of MSD for different water models at 298 K